

DIMPLE SHARMA

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SUMMARY

AI/ML engineering intern and final-year B.Tech (AI & ML) student with hands-on experience in machine learning, deep learning, computer vision and Generative AI (RAG, LLMs). Builds end-to-end AI applications with Python, FastAPI, Node.js/TypeScript, React and PostgreSQL, including JWT authentication, role-based access control, automated testing and AWS EC2 deployment.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

AI / Machine Learning: Machine Learning, Deep Learning, Computer Vision, PyTorch, scikit-learn, XGBoost, YOLOv8, OpenCV

Generative AI: LLMs (Gemini), Retrieval-Augmented Generation (RAG), embeddings, FAISS, BM25 hybrid search, OCR (Tesseract)

Backend: Node.js, Express, FastAPI, REST APIs, JWT authentication, Role-Based Access Control (RBAC), bcrypt

Databases & Data: PostgreSQL, SQL migrations, Pandas, NumPy, data preprocessing, feature engineering, model evaluation

Frontend: React, Vite, Tailwind CSS, Streamlit, Figma

Cloud & Tools: AWS EC2, Application Deployment, Git, GitHub, pytest, Node.js test runner

EXPERIENCE

AI & Agentic Automation Intern | PanScience Innovation, New Delhi, India

06/2026 – Present

- Contributed to an AI-powered tender and procurement automation platform (FastAPI, Next.js, Gemini), integrating document processing, RAG-based retrieval and LLM document analysis into tender workflows.
- Implemented OCR text extraction for scanned procurement documents (Tesseract, PyMuPDF) in English and Hindi.
- Worked on role-based access control with JWT authentication, defining permission boundaries for Procurement Officer, Evaluator, Approver and Administrator roles.
- Completed a data annotation pilot for a US-based client (via Indika AI): prepared the initial report, then annotated 9 pilot videos per the client's report and delivered structured annotation and metadata CSVs.

PROJECTS

RailTel Procurement Chatbot (RAG) | *Node.js, TypeScript, Express, FastAPI, PostgreSQL, FAISS, Gemini, React*

- Built a permission-aware RAG chatbot: a Node.js/TypeScript (Express) REST API on a PostgreSQL schema with SQL migrations, plus a Python FastAPI retrieval service using Gemini embeddings and FAISS with BM25 hybrid search.
- Implemented JWT authentication with bcrypt and RBAC over a public and three department knowledge bases, enforcing permissions before retrieval and again whenever a cited source is opened.
- Developed PDF/DOCX/HTML ingestion with Tesseract OCR, heading-aware chunking and robots.txt-compliant website crawling, plus admin tools for document versioning, indexing and user management.
- Evaluated answers on a 50-query benchmark of real procurement questions (38/50 fully correct); diagnosed failures and added lexical recall and coverage-aware context selection, recovering the missing evidence for 5 of 7 failed queries in offline replay.
- Delivered source-cited answers in a React/Vite UI, backed by 74 Node.js and 23 pytest test files.
- Deployed the application (Node.js API, FastAPI RAG service, React frontend) on a provided AWS EC2 instance.

PredCo Sentinel – CCTV Video Analytics (Prototype) | *Python, FastAPI, OpenCV, YOLOv8, React, Tailwind CSS*

- Built a FastAPI and React/Tailwind CCTV dashboard that ingests a live RTSP stream via OpenCV (threaded capture, reconnect backoff), runs YOLOv8n object detection and streams annotated video.
- Implemented detection-triggered alerts with severity levels, annotated snapshots and a triage workflow (new, acknowledged, under review, resolved), plus rule-based natural-language event search.
- Built a paginated 30-camera HLS monitoring wall via a backend proxy with per-feed status.

Smart Recycling AI – Waste Classification | *PyTorch, Streamlit*

github.com/Dimple-Proj/Smart-Recycling-AI

- Fine-tuned a pretrained ResNet18 to classify six waste categories with data augmentation, benchmarked against a custom CNN; built a Streamlit app showing top-3 predictions and recycling guidance.

Credit Risk Prediction | *XGBoost, scikit-learn, SHAP, FastAPI*

github.com/Dimple-Proj/credit-risk-ml

- Built an XGBoost loan-default model on the Home Credit dataset in a scikit-learn pipeline with class-imbalance handling, GridSearchCV tuning and 5-fold ROC-AUC validation; explained with SHAP, served via FastAPI.

EDUCATION

B.Tech, Artificial Intelligence and Machine Learning | JEMTEC, Greater Noida, India

Expected 2027

CGPA: 9.0 (overall) | Coursework: Data Structures and Algorithms, Database Management Systems